

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Examination, 2019
CE 3107
(Elements of Solid Mechanics)

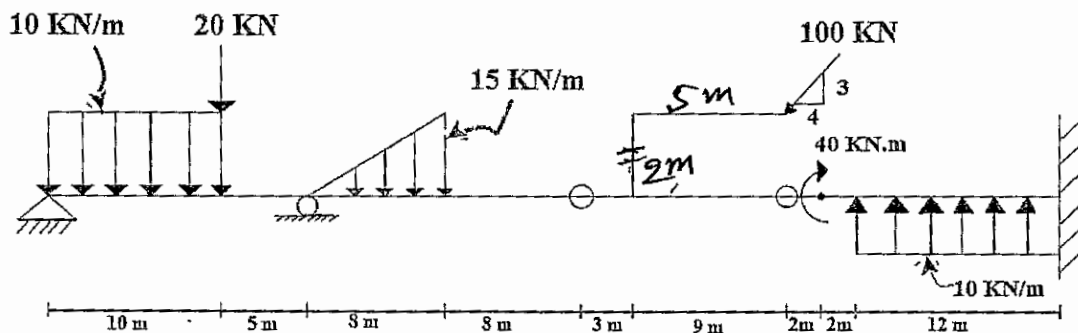
Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Assume reasonable value for missing data.
iii) Answer any three questions from each section in separate script.

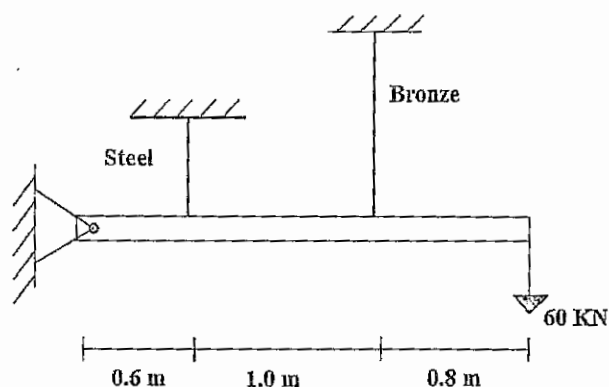
Section – A

1. (a) Define shear force and bending moment. Show the relationship among load, shear and moment. (10)
- (b) Draw the shear force and bending moment diagrams for the beam as shown in figure below: (25)



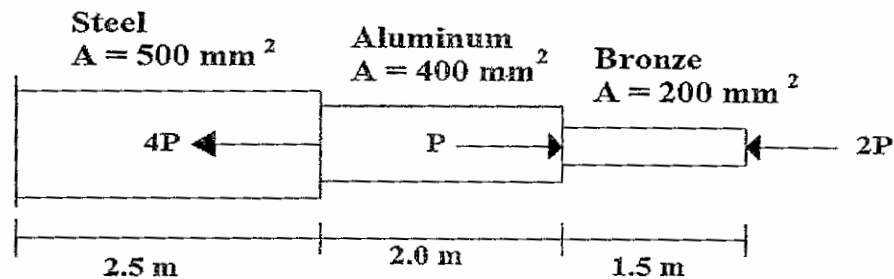
2. (a) Draw a typical stress strain diagram for mild steel. If a well-defined yield point is not found for a material, how can you determine yield strength? – describe. (10)
- (b) Obtain an expression for the elongation ' δ ' of a vertical bar of length ' L ', uniform cross sectional area of ' A ', supported at its upper ends under its own weight ' W '. (10)
- (c) A horizontal bar of negligible mass, hinged at A in the following figure and assumed rigid, is supported by a bronze rod 2.0 m long and a steel rod 1.0 m long. Using the data in the accompanying table compute the stress in each rod. (15)

	Bronze	Steel
Area (mm ²)	300	600
E (GPa)	83	200
Proportional Limit (MPa)	140	240

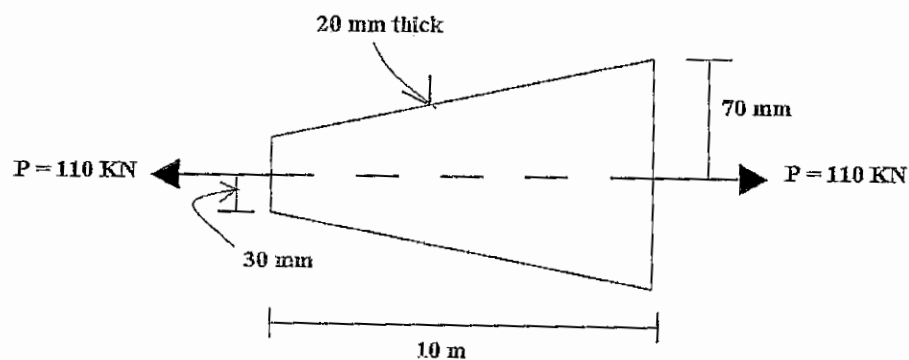


3. (a) Define simple stress. Prove that uniform stress distribution can exist only if the resultant of the applied loads passes through the centroid of the cross section. (08)

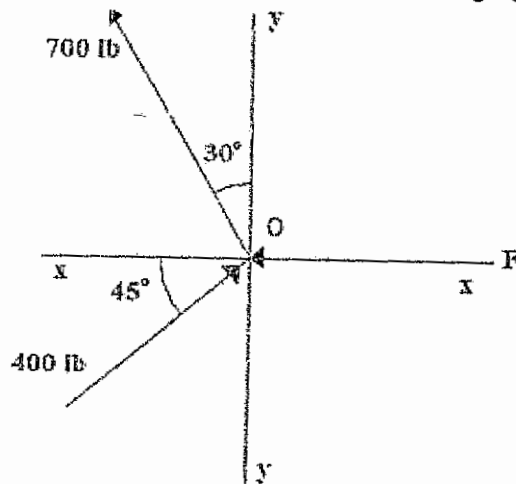
- (b) An aluminum rod is rigidly attached between a steel rod and a bronze rod as shown in figure below. Axial loads are applied at the positions indicated. Find the maximum value of P that will not exceed a stress in the steel of 140 MPa, in aluminum of 90 MPa, or in bronze of 100 MPa. (12)



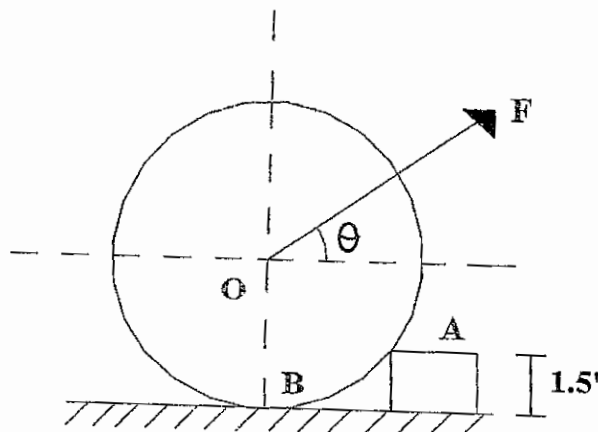
- (c) Compute the total elongation caused by an axial load of 110 kN applied to a flat bar 20 mm thick, tapering from a width of 140 mm to 60 mm in a length of 10 m as shown in the following figure. Assume $E = 200$ GPa. (15)



4. (a) Determine the resultant of the forces shown in the following figure if $F = 500$ lb. (12)

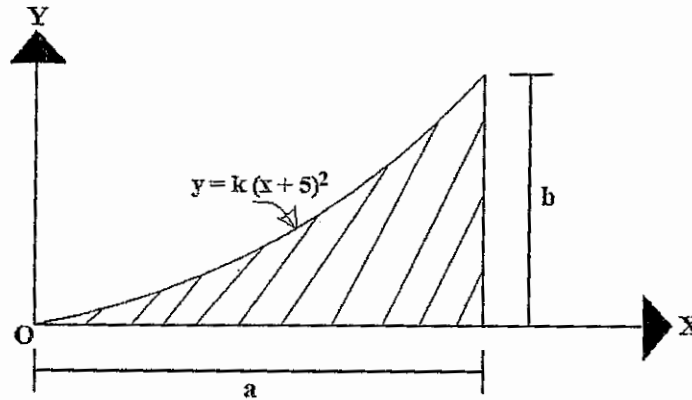


- (b) A 2500 lb wheel with a radius of 3 ft is acted upon by a force F in the following figure, which ends to pull the wheel over the obstruction at A. At the instant the wheel is about to move, the pressure between the wheel and the ground is zero. What is the magnitude of the force F at this instant, if $\theta = 30^\circ$? (23)

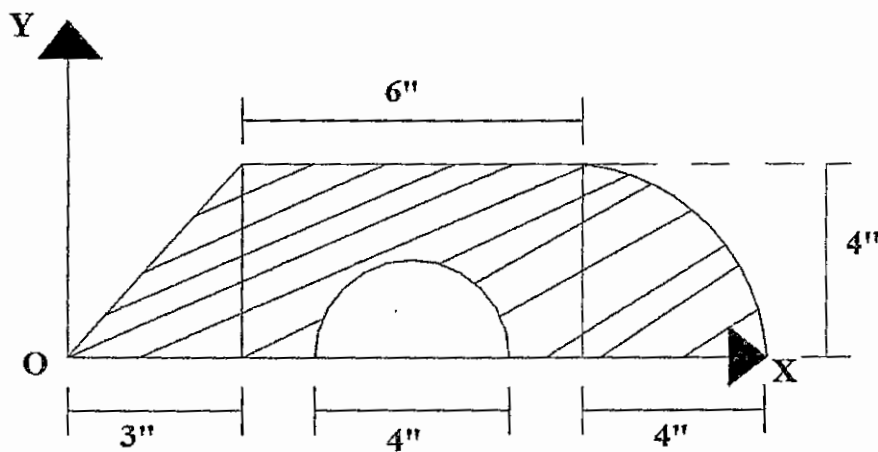


Section – B

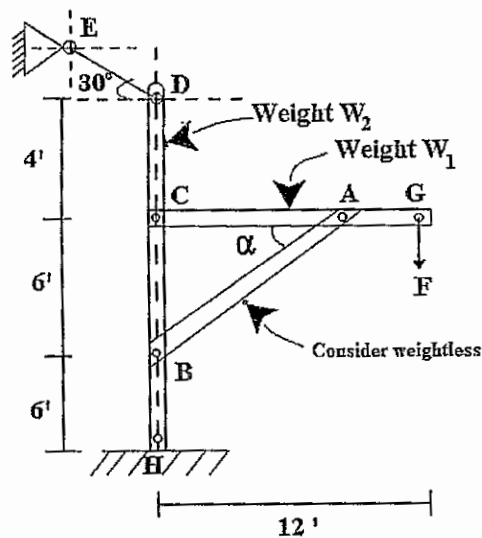
5. (a) What is center of gravity and centroid of an area? Derive the formula for centroid of lines. (10)
- (b) By direct integration, determine the location of centroid of a parabolic spandrel in both directions. (10)



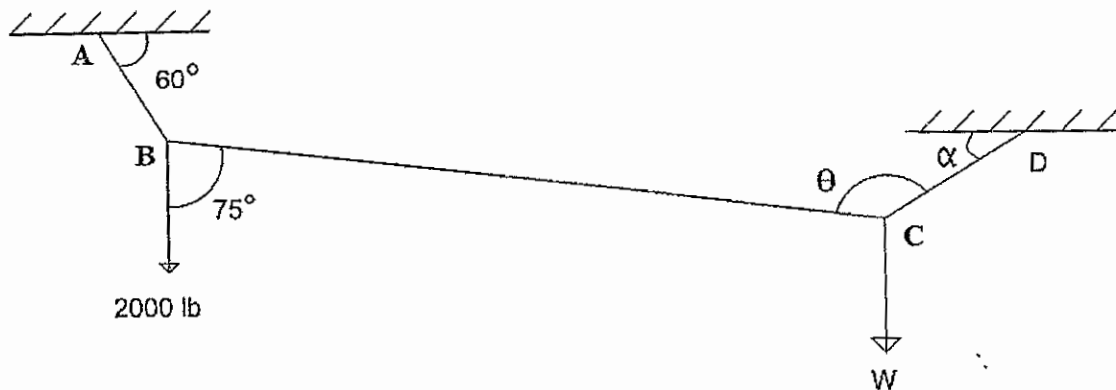
- (c) Determine radius of gyration and polar moment of inertia with respect to the centroidal axis and XY axis about the point O. (15)



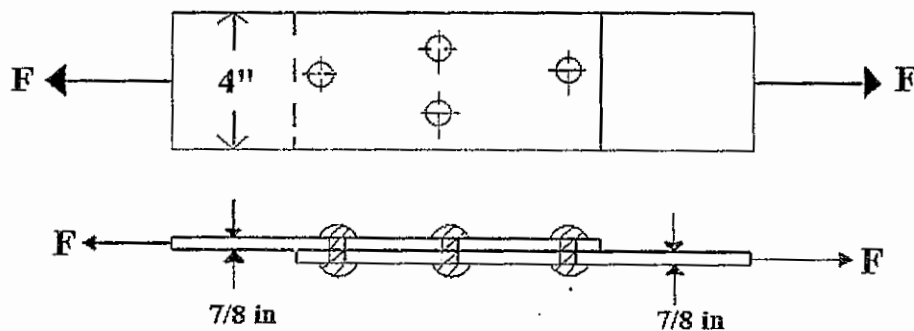
6. (a) Define the following terms: (09)
- (i) Transmissibility of Force (ii) Free body diagram (iii) Concurrent force system
- (b) Draw the free body diagram of different members in the following figure: (11)



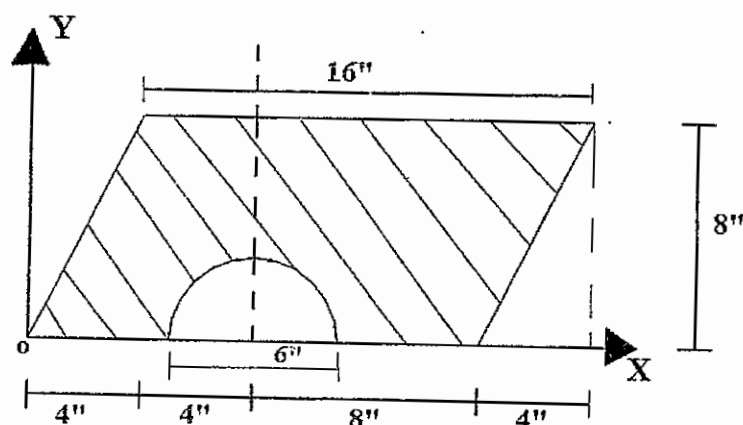
- (c) The weights are suspended from a flexible cable as shown in figure below. For $\theta = 120^\circ$, determine the internal forces in the various parts of the cable and weight W . (15)



7. (a) Define the following terms: (10)
 (i) Normal stress (ii) Bearing stress (iii) Normal strain (iv) Poisson's ratio
 (v) Tangential stress.
- (b) A hollow steel tube with an inside diameter of 200 mm must carry a tensile load of 800 kN. Determine the outside radius of the tube if the stress is limited to 240 MN/m². (10)
- (c) The lap joint shown in the following figure is fastened by four 3/8 in diameter rivets. Calculate the maximum safe load F that can be applied if the shearing stress in the rivets is limited to 28 ksi and the bearing stress in the plate is limited to 36 ksi. Assume the applied load is uniformly distributed among the four rivets. Also determine the normal stress of the plate. (15)



8. (a) Define moment of inertia. Prove that $I_x = \bar{I}_x + Ad^2$ where the symbols have their usual meaning. (12)
- (b) For the shaded area of the following figure determine: (23)
 (i) I_x (ii) The moment of inertia about centroidal axis parallel to X axis. and (iii) Polar moment of inertia for an axis through 'O'



Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 3rd Year 1st Term Examination, 2019

CE 3151

(Fundamentals of Transportation Engineering)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) What is a transportation system? Describe different modes of a transportation system. (10)
(b) What are the benefits of a good system of roads? (10)
(c) Describe the interdependence of land use and transport planning. Explain the stages of transportation planning. (15)
2. (a) What are the criteria for selecting an external cordon line? What is zoning? (10)
(b) What is traffic volume? Define the following terms: (10)
(i) AADT (ii) AAWT (iii) ADT (iv) AWT
(c) What is O-D survey? What are the objectives of O-D survey? Describe the common methods of collecting O-D data. (15)
3. (a) What are the driver's age and turnover according to the act of EU-27? How can employment survey be conducted? (10)
(b) What does infrastructure include? Distinguish between the terms of economic infrastructure and social infrastructure. (10)
(c) What is the role of transportation in the global supply chain? What are the conditions under which rail transport is undertaken? What are the advantages and disadvantages of intermodal transportation system? (15)
4. (a) What types of studies are carried out while planning a highway system for a particular area? (12)
(b) What is transportation planning? How results are interpreted from the data collected during carrying out road planning studies. (11)
(c) What are the factors which affect the road alignment? Discuss five of the factors in details. (12)

Section – B

5. (a) Classify roads according to their functions and explain them briefly. (10)
(b) Define: (i) Kerb; (ii) Carriageway width; (iii) Park and walk; (iv) Park and ride (08)
(c) What is parking study? Write down the different types of parking facilities. (10)

- (d) Write down the locations where parking should be prohibited. (07)
6. (a) What is meant by 'Geometric design of highways'? Draw a neat sketch of a typical rural and urban roadway section with its major elements. (10)
- (b) Explain the role of road surface characteristics in highway geometric design. (06)
- Write down the differences between skid and slip.
- (c) Why is extra-widening provided on the horizontal curve? (07)
- (d) A concrete road has 12m width. Design with dimensional sketch a composite chamber for the given road section. Assume the ordinate at least 1m interval. (12)
7. (a) Define: (i) Speed; (ii) Flow; (iii) Concentration (06)
- (b) What is bottleneck? Write down the causes of bottleneck. (05)
- (c) Write down the causes of traffic congestions. (12)
- (d) Write down the traffic engineering tools for good traffic systems. (12)
8. (a) What is camber? Why is it provided on roadway? What are the recommended values of camber for different types of road surfacing given by I.R.C.? (10)
- (b) What is super elevation? Derive an expression for super elevation at level road. (10)
- (c) Calculate the stopping sight distances on a graded highway for a design speed of 80 km/hr. Assume, Reaction time 2s and a value of μ is 0.35 (15)
- i. When grade is 3% descending
- ii. When grade is 3% ascending
- iii. When road is flat

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 3rd Year 1st Term Examination, 2019

Hum 3117

(Accounting and Auditing for Planners)

Full Marks: 210

Time: 3 hrs

- N.B.**
- i) Answer any three questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.
 - iii) Assume reasonable value for any missing data.

Section – A

1. (a) Define accounting. (05)
- (b) Who are the users of accounting information? Discuss. (10)
- (c) On June 1, 2018 Mr. Amin established the Dhaka Travel Agency. The following transactions were completed during the month: (20)

2018	
June 1	Invested Tk. 200,000 cash in the agency
June 2	Paid Tk. 5000 cash for June office rent
June 3	Purchased office equipment for Tk. 25000 cash
June 4	Incurred advertising expenses of Tk. 3000 in the "Daily Desk" on account
June 5	Paid Tk. 5000 cash for office supplies
June 6	Earned Tk. 120,000 for service rendered: cash Tk. 30,000 is received from customers and the balance of Tk. 90,000 is billed to customers on account.
June 7	Paid the "daily desk" amount due in transaction (4)
June 8	Withdrew cash Tk. 4000 for personal use
June 9	Paid salaries to employees Tk. 15,000
June 10	Received Tk. 90,000 in cash from customers who have previously been billed in transaction June (6)

Required: Journalized above the transactions.

2. (a) What is meant by journal? (05)
- (b) Distinguish between journal and ledger. (05)
- (c) On 1st January, 2018 Mr. Hafiz started a firm. The following transactions were completed during the month: (25)

2018	
January 1	Invested Tk. 200,000 cash to start the firm
January 2	Paid Tk. 800 cash for January office rent
January 3	Purchased office equipment for Tk. 3500 cash
January 4	Incurred Tk. 600 for advertising cost in the "Daily Star" on account
January 5	Paid Tk. 1000 cash for office supplies
January 6	Earned Tk. 12,000 for service rendered: cash Tk. 4000 received from customers and the balance of Tk. 8000 is billed to customers on account.
January 7	Withdrew cash for personal use Tk. 200
January 8	Paid the "daily star" due in transaction (4)
January 9	Paid employee salaries Tk. 2100
January 10	Received Tk. 8000 in cash from customers who have previously been billed in transaction (6).

Required: Prepare a tabular analysis of the transactions using the following column headings: cash; Accounts Receivable; Supplies; Office Equipment; Accounts Payable and Hafiz's Capital

3. (a) What is trial balance? What are the limitations of a trial balance? (08)
- (b) What is adjusting entries? Discuss the objectives of adjustments. (07)

- (c) From the following ledges balances of Luna company prepare a trial balance as on December 31, 2017: (20)

	Taka		Taka
Cash.....	10,940	Equipment.....	14,900
Accounts Receivable.....	9780	Accumulated depreciation...	5400
Accounts payable.....	6220	Rent income.....	6500
Unearned revenue.....	1800	Depreciation expense.....	4000
Luna's capital.....	45,200	Salaries expense.....	56,700
Luna's drawing.....	15,000	Utility expense.....	16,900
Commission revenue.....	63,100		

4.

Mr. Kalam
Trial Balance
As on December 31, 2018

(35)

Account Titles	Debit (Tk.)	Credit (Tk.)
Cash.....	1750	
Accounts Receivable.....	3109	
Office supplies.....	382	
Office Equipment.....	3755	
Accounts payable.....		796
Unearned revenue.....		5000
Kalam's capital		4000
Kalam's withdrawals.....	6000	
Legal fees.....		17,575
Rent expense.....	1800	
Utility expense.....	717	
Wages expense.....	9858	
	27,371	27,371

Adjustments: (a). Unused supplies during the year Tk. 182.
(b). Estimated depreciation on equipment Tk. 600
(c). Wages unpaid Tk. 142

Required: a) Prepare a statement of comprehensive income for the year ended December 31, 2018.

b). Prepare owner's equity statement and

c). prepare a statement of financial position as on December 31, 2018.

Section – B

5. (a) Define audit. (05)
(b) Describe the primary objectives of audit. (10)
(c) What is meant by misappropriation of asset? (05)
(d) Differentiate between internal and external audit. (15)
6. (a) What is social audit? (05)
(b) Classify audit in terms of legal requirement. (10)
(c) Differentiate between financial statement audit and compliance audit. (15)
(d) What is meant by "inherent risk"? (05)
7. (a) Define tax and tax holiday. (08)
(b) What are the conditions to be fulfilled to get tax holiday by an entity? (12)
(c) Describe any six (6) principles of taxation. (15)
8. (a) Define internal control. (05)
(b) Discuss the components of internal control. (15)
(c) What are the essential characteristics of internal control system? (15)

Khulna University of Engineering & Technology
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BURP 3rd Year 1st Term Examination, 2019.

URP 3161
(Housing and Real Estate Development)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What do you mean by 'Housing Right'? Explain the five essential elements of functional housing market. (10)
(b) One continuing problem in Bangladesh is "affordability" in housing sector. What will be your suggestion for future housing policy to meet up the challenges? (10)
(c) Recall a segment of your life where you feel the deficiency of adequate housing. How can you solve the problems? (07)
(d) How can you delineate the formal and informal housing? Slum and Squatter – what is the difference? (08)
2. (a) Today's world is for the wealthy people. How can you relate it with the terms "insurgence" and "informality"? (15)
(b) The economic crisis from 1998 to 2000s saw a decline in physical and economic development in Jakarta-Bandung (JBR). Briefly discuss the scenario. (15)
(c) What is the difference between condominium and apartment? (05)
3. (a) Why 1990 is important in housing concept? Mention subsequent international housing policy changes in the past in housing sector. (15)
(b) What are the major component of the Fifth Five Year Plan that deals with the current urbanization trend and housing demand in the urban area? What are the major strategies for the provision of housing for the poor? (10)
(c) Singapore is considered as a success story of public housing scheme. Explain the major components of that housing scheme. (10)
4. (a) Capitalistic private sectors are more efficient rather than socialist one. What do you think? (10)
(b) What are the major sources of different housing sectors in Bangladesh? (07)
(c) Suppose you are assigned to calculate the tax of urban housing property. What will be your considerations? According to Wendt's model calculate the land value if Net income = 100000 and CR = 350. (08)
(d) Write short notes on the followings: (10)
i. Role of BRAC in housing sector.
ii. Land Banking.

Section – B

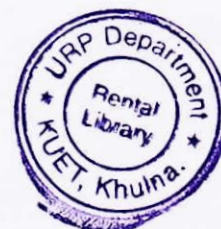
5. (a) Is real estate a property? (05)
(b) "Real estate includes not only the land surface of the earth but also the permanent improvements above and below the surface."-Why? Explain with example. (15)
(c) Real estate market tends to be highly segmented and localized -Why? Explain with relevant examples. (15)
6. You have been appointed as an intern in East West Property Development Ltd (EWPD) of Basundhara Group. You are assigned for the following assignment. The Padma Bridge will connect Louhajong, Munshiganj to Shariatpur and Madaripur, linking the south-west of the country, to northern and eastern regions. The company plans to invest on raw land near the Padma Bridge. Now you have to do the following tasks.
(a) Task 1: Explain the nonmonetary effects of that investment value. (10)
(b) Task 2: Identify the possible investment risks. Why investment in that raw land is highly risky? Explain in terms of cash flow uncertainty. (15)
(c) Task 3: Your Company will purchase a parcel of land in next month for BDT 750 million. For how much will your company have to sell the property in 15 years to earn a 10% annual return on both initial outlay and expected annual payment of 8.5 million for property tax and development to the land? Assume these funds could be invested at comparable risk to earn a 10 percent annual return. (10)
7. (a) Discuss different types of property cycles in real estate market. Clarify the anatomy of a real estate cycle with example in context of Bangladesh. (10)
(b) Market research approach focuses from broad to specific. Explain with example. (10)
(c) When will the downturn actually occur in a real estate market? Explain with supply and demand curve. How a developer will survive during a market downturn? (15)
8. (a) What is real estate valuation? Summarize the primary principles of real estate valuation. (20)
(b) What are the adjustments need to be considered for valuation of property using the traditional sales comparison approach? (15)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 3rd Year 1st Term Regular Examination, 2019

URP 3121

(Rural Development Planning- I)



Full Marks: 210

Time: 3.00 hrs

- N.B.**
- i) Answer any three questions from each section in separate script.
 - ii) Figures in the right margin indicate full marks.
 - iii) Assume reasonable value for any missing data.

Section – A

1. (a) Describe the application of Zubin and Spring's model of stress /vulnerability in urban life .Does it similarly applicable for rural life? (12)
- (b) Compare the significance of policy talk and technical talk in rural development plans / projects. (15)
- (c) Write short notes on: (08)
 - (i) Economic Growth and Economic Development
 - (ii) Leading and Lagging Industries
2. (a) "Urban trap is limiting rural development"- Do you agree? Justify your answer. (10)
- (b) Mention some biological, psychological and social stressors which make urban life stressed than rural life. (15)
- (c) Define Rural Access Index (RAI). Give an example of RAI application in rural development. (10)
3. (a) "Squared Poverty Gap Index (SPGI) takes into account inequality among the poor ". Define this sentence with an example. (08)
- (b) Determine (i) Head Count Ratio (ii) Poverty Gap Index (iii) Squared Poverty Gap Index for the following countries and explain the results. Assume poverty line is 110000. (15)

Expenditure for each individuals in countries				
X	30,000	11,5000	11,0000	90,000
Y	45,000	50,000	95,000	25,000
Z	15,0000	80,000	13,0000	15,0000

- (c) What are the key principles of World Happiness Index? How they are related to rural development? (12)
4. (a) "Lorenz Curve and Gini Coefficient are best suited to measure relative poverty". Describe this statement with income equality and inequality examples. (12)
- (b) Mention each steps of HDI calculation with appropriate example and also give its importance. (13)
- (c) Describe Todaro's development objectives with real life examples. (10)

Section – B

5. (a) Write about the basic four principles of defining a rural area. How will you demarcate a rural area following the principles? (07)
- (b) What are the common forms of "Scattered Settlement" found in rural areas? Give necessary sketches. (10)
- (c) "The gap between urban and rural divide is being narrowed down with the growth of modern communication and technology". Do you agree with the statement? – Explain. (10)
- (d) "Different factors influence rural settlement pattern" – Justify the statement. (08)
6. (a) "Highest level growth foci require higher threshold population." Explain the statement with appropriate examples. (10)
- (b) Briefly describe the best method for determining the present centrality of settlements for small area? Use the example of health service in this regard. (15)
- (c) How do you identify suitable industries in a rural area by using industrial development screening matrix? (10)
7. (a) Which types of houses are normally found in hoar basin areas and along major rivers of Bangladesh? (05)
- (b) Describe the criticisms of the model for institution building of rural development. (10)
- (c) Which component of Comilla Experiment reflects decentralized and coordinated rural administration? Explain with necessary example. (10)
- (d) Briefly discuss the impact of "One House, One Farm" Programme on Social and Economic Development in rural areas of Bangladesh. (10)
8. Write short note from the followings (any five): (35)
- a) FFWP
 - b) Timber Houses in Bangladesh
 - c) Spatial planning for Rural Areas
 - d) Success of V- AID programme of rural development
 - e) Governmental Organizations involved in Development Activities.
 - f) Cruciform Settlement